

Varen 6869 AgSp1.1 cr	PGNNPGVEMGPDGQINKIILPSTPANPSSSGNPNG..Q.....	36
Varen 6869 AgSp1.2 cr	PGNTPGAELGPDGQINRIILPSTPPNPSSGGLNPDGEP.SAPFGPGNSFNSPK.....SPGFKFPGFPGAPGSMGPGVGMDFSQLPSGESPE	87
Mg TRINITY DN1546 c0 g1 Cr	GGNTPGAELDPDGQIKKVILPSTPSEPFSSSSSLPSQ.....DSQLNGREPYE	47
Msag 7048 AgSp1 cr	PGHTPGAELGPNNGNINRIFLPTAPNPPSPGGLNPPGEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPFGFPGAPGNSGPIGYIDFSQLPNGQPPE	93
Ncruc 58532and13532 AgSp1 cr	PGNTPGAELMGPDGHHINRIVLPTPSNPQSSSGPQYPGGGGPFGPGNSPNSPGQSPGNYPGSAFQFPFGFPGAPGNSGPIGYIDFSQLPNSGESPE	95
Msag 19329 AgSp1.1 cr	PGTTPGVVTGSDG.....QPVK.....Y	18
Msag 18528 AgSp1.2 cr	PGTTPGVRIIDPDGKINNVLVLPISPNSELGGLNPDNEP.IPPFGPDNTPNSP.NFPGNYPRSTFKFPGFPGAPGSMGPIGFLDFSELS..ESPE	91
Lcorn 51676 AgSp1.1Cfrag cr	PGNTPGAELGPDGQITRIILPTPPPNNPSSGGLNPAAGEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPFGFPGAPGNSGPIGYLDFSQPLPSDVSP	93
Lcorn 58650 AgSp1.2 cr	PGNTPGAELGPDGQITRIILPTPPPNNPSSGGLNPAAGEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPFGFPGAPGNSGPIGYLDFSQPLPSDVSP	93
Amarm 50219 AgSp1.1 cr	PGNTPGAQLGPDGHHINRIVLPTAPNPPSSGGLNPAAGEP.TAPFGPGNSPNSP.QSPGNYPGSAFKFPGFPGAPGANGPIGYIDFSQLPSGESPE	93
Aarg AgSp1 FC511 cr	PGTTPGAELGPDGKINKIVLPTTPPNPPPSGGLNPNENEP.IPPFGPGNSPNSP.QTPGNYPGSAFQFPFGFPGAPGNSGPIGYIDFSQLPSGESPE	93
Atri AgSp1 genomic cr	PGTTPGAELGPDGKINKIILPTTPPNPPSSGGLNPKGEP.IPPFGPGNSPNSP.QPPGNYPGSAFQFPFGFPGAPGANGPIGYIDFSQLPSGESPE	93
Mhut 1271 AgSp1 cr	PGITPGVELGPDGHHINRIVLPTPPPNNLFLSDPLNPVKGK.AAPFGPGNIPNSP.YSPGNYPGSAFQFPFGFPGAPGNSGPIGYIDFKQLPSGVSPE	93

Varen 6869 AgSp1.1 cr	SFRPSGDPFSNLGPDGSLNLTDSIPDFDNPGYGSFGYPQVPLGYFDSVP.....NSDPSSQFVLPGSPGSPGSDGQFPFG	109
Varen 6869 AgSp1.2 cr	MYGSIGFLPNFSPEIGGPFPGFPGGGGDFGPGGYLNTDSLPDFINPGYGFPGSPQAPLGDFDNKLPDFNSDFSGMFVLPGSPGSPGSGNGQFPFG	182
Mg TRINITY DN1546 c0 g1 Cr	MDDSVGHFNPFNLEIGGSFLGYEGGPNNLGLGENLNIEITLPYFIKPGSGFGGSSHVPFGHFDYNQLPPDFNPNFPGLFALPGYSGFPESSRSQFAG	142
Msag 7048 AgSp1 cr	QVGSIGYFDFSGEIGGPFPGFSGGGGGPGGCGYLNIMSLPDTNRYGYGFPGSPQAPLGYLDESMLPKGFNPNFSGGFVYPGYPGSPGSGGGQYYPG	188
Ncruc 58532and13532 AgSp1 cr	MVGNIGFFPNFSPEIGGPFPGFPGGGDDSGPGGFLNIGSLPNFKSPGYGFPGSPQAPLGYLQNMMLPPGYNPDFSGLPLVFPGYSGSPGSGGQLPG	190
Msag 19329 AgSp1.1 cr	IAPPGGISPNFGLEIGGVFPSSISAGADDSAPGEHLNIHSIPNLINPLYGFPSFQAPLGIFYSSQ.....SSEFSDKFVLPGDLLSPGSEGGQFLG	108
Msag 18528 AgSp1.2 cr	FDGSIGFFPNFNPFEIGGPFPGFSGGGGEDSGPGGYLDIDSIPDFINPLYGFPGSPQAPLGFIHNP.....SSDVSDTFVLPGDPCAPGSQCGFPFG	181
Lcorn 51676 AgSp1.1Cfrag cr	MVGPPIGFFPNFNPFEIGGQFPFGFPGGGDDSGPGGFLNINSLPDFRNPYGFPGSPQAPLGYLDESMLPQDFDFDFSGGFVFPFGYPGSPGSGSQFPFG	188
Lcorn 58650 AgSp1.2 cr	MVGPPIGFFPNFNPFEIGGQFPFGFPGGGDDSGPGGFLNINSLPDFMNPYGFPGSPQAPLGYLDESMLPQDFSLDFSGGFVFPFGYPGSPGSGSQFPFG	188
Amarm 50219 AgSp1.1 cr	MVGSIGFFPNFSPEIGGPFPGFSGGGGNSGPGGFLNINSLPDFMNPVYGFPGSPQAPLGYLKSMPLPGFNLDSSGQLVFPDGPSPGSGSQFPFG	188
Aarg AgSp1 FC511 cr	MEGSIGFLPNFSPEIGGPFPGFPGGGGSPGLSGFLNVNLSLPDFINQGYGFPGSPQDPLGYLNFMSMLPPGYNPDFSQGFVYPGYPGSPGSGGQFPFG	188
Atri AgSp1 genomic cr	MEGPPIGFLPNFSPEIGGPFPGFPGGGGSSGPGGFLNVNLSLPDFINQGYGFPGSPQDPLGYLNFMSMLPPGYSPDFSQGFVYPGYPGSPGSGGQFPFG	188
Mhut 1271 AgSp1 cr	MIGSIGFLPNFSPEIGGPFPGFPGGGDDSGPGGYLNIDSLPDFNKPQGYGFPGSPQAPLGYLKSKLPPGYNPFGSQGLVFPFGYPGAPGSGSQFPFG	188

Varen 6869 AgSp1.1 cr	GFLSFMQLPKSITDKINRTFDFTELIQAVQPLPFGNTINLGAIPRDNLQDIPGFDGSYNNLKFDYFGDSTKQNGGIFYIPELIIYLINQLPG....	200
Varen 6869 AgSp1.2 cr	GFLSFPQLPKDITDKIDRPFSFPQLIQAIIQPLPFGDSINVGAIPEDKDKDMPGFNGTSDSLRLDNFEDNNRPSGGVFYIPELVRLISYLPVGSFG	277
Mg TRINITY DN1546 c0 g1 Cr	GFLSFPQLPKGITNNINGSFSFAELIQALQPLPFGKSINVGAIPQNKLVMPGFNGTYDNLGLDIDGSSHPNGGVFYLPELVRLVGYLPVGSFG	237
Msag 7048 AgSp1 cr	GFLSFSQLPESITNIHGSFSFSQMMNALGPLPFGQ.VNIGAIPKNQLQSIIPGFNGNYDNLMLDNYGDSSQPTGGVFYLPPELVRLIASYLPESFS	282
Ncruc 58532and13532 AgSp1 cr	GFLSFPQLPKDVTDKISSFTSFPELLKSLQPLPFGQINMGAIPQDKLQNIIPGFNGTYDNLKLANYGDNQPTGGVFYLPPELVRLMSYLPVGSFG	285
Msag 19329 AgSp1.1 cr	GFLSFPQLSKDITNLINGSFSFAELIQALQPLPFGNSINVGAIPHSELVMPGFNGTYDNLKLDNIKNNGHPNGGVFYLPELVRLIDFVPVGSFN	203
Msag 18528 AgSp1.2 cr	GFLSFPQLPKSITDMINGTFSFQELIQALQPLPFGNSINVGAIPDDKLQDIPGLKGNHDSLKLSQLGDNSQPSGSLFYIPELMILISHLPD....	272
Lcorn 51676 AgSp1.1Cfrag cr	GFLSFPQLPKDITDKINGTFSFPQLINALQPLPFGQINIGAIPQDKLQNIIPGFNGSYDNLKLANFGDNSQPTGGVFYLPPELVRLISYLPVGSFG	283
Lcorn 58650 AgSp1.2 cr	GFLSFPQLPKDITDKISGTFSFPQLINALQPLPFGQINIGAIPQDKLQNIIPGFNGSYDNLKLANFGDNSQPTGGVFYLPPELVRLISYLPVGSFG	283
Amarm 50219 AgSp1.1 cr	GFLSFPQLPKIITDKIDGTFSFPQLIEALQPLPFGQINIGAIPKDKLQDIPGFEGTYDNLKLANFGGNNQPTGGVFYLPPELVRLIGYLPVGSFG	283
Aarg AgSp1 FC511 cr	GFLMSFPELPKDVTDKINGTFSLPQLMQALQPLPFGQINIGAIPKQMQNIIPGFEGTYDNLRLANFGNNNQPVGGVFYLPPELVRLISYLPVGSFG	283
Atri AgSp1 genomic cr	GFLSFSFELPKGVITDKINGTFSLPQLMQALQPLPFGQINIGAIPKDKMQNIIPGFEGTYDNLRLANFGSDNQPTGGVFYLPPELVRLISYLPVGSFG	283
Mhut 1271 AgSp1 cr	GFLSFPQLPKDITDKISGTFSFPQLINSLQPLPFGQINIGTIPQDKLQDIPGFTGTYNLKLKLANFGDNSQPTGGVFYLPPELVRLIGYLPVSSS	283

Varen 6869 AgSp1.1 cr	GISGT F DFPGFYGPRRN I CDAG N ...D S SDSGSQDLGGEVE G DL D GLFGD M GDAAPSDT M DAAPR.....	263
Varen 6869 AgSp1.2 cr	K GS G TINQDGG F CEPFNFPG L NGAPGN I CDYPEDYYDY S MDEGSQDLGGEVE G DGNGPSGD V GDAAPED I GAAP P NNLSP.....	358
Mg TRINITY DN1546 c0 g1 Cr	I GP G TINQ E GGFGDAIK F PGLNGAPGYICDYP E NG.. D ATDRGSQDLGEGV F GDGNTD I NDD G FQES.....	302
Mraby 7048 AgSp1 cr	 A NGGFGDP F GE Q LLHGAPGYICDYPDND.. D TTAGGSQDLGGE I KGNHGPSGDVADAAP.. N LGS S APGNDLGGAAPG.....NT	359
Ncruc 58532and13532 AgSp1 cr	K GP G TIN H DGG F DDPFNFPG L NGAPGS I CDYPDNG.. D ATDGGSQDLGGEV K GDNGP A GDVADAAP.. D L G APAPGNDLGGAAPDLGAPAPGND	376
Msag 19329 AgSp1.1 cr	K GL G TINQ E GGFG H PFNFPG L SGTLGYICD L ENG.. D ATIEGSQIFS.....	249
Msag 18528 AgSp1.2 cr	I GP E TIN S DGG F STPFNFPG F SGPPGYICDYPD K D.. D VTDGGF H DLGG V KGN D NR P FGDV G GAAPDEDD A NDVAS Q DLGGEV K .INKTGPFRD	364
Lcorn 51676 AgSp1.1Cfrag cr	K GS G TINQDGGFGDPFNFPG L NGAPGYICDYPDNG.. D ATAGGSQDLGGEVKGNGNGPSGDVADAAP.. D LGSPAPENDLGGAAP.....	364
Lcorn 58650 AgSp1.2 cr	K GS G TINQDGGFGDPFNFPG L NGAPGYICDYPDNG.. D ATAGGSQDLGGEVKGNGNGPSGDVADAAP.. D LGSPAPENDLGGAAP.....	364
Amarm 50219 AgSp1.1 cr	Y GP G TIN H DGGFG H PFNFPG L NGAPGYICDYP N NG.. V L...SQDLGGEVKGNGNGPSGDVADAAPENDLG G TAPEND I GGAAP EN.	364
Aarg AgSp1 FC511 cr	N GP G TINQDGGFGDPFNFPG L NGAPGYICDYPDNG.. D AAT G RSQ N L G GE I KGNNGNGPSGDVADAAP E ADLGGAAPQSD L A.....	362
Atri AgSp1 genomic cr	N GP G TINQ N GGFGDPFNFPG L NGAPGYICDYPDNG.. H ATTGGSQ N L G GE I KGN D NGPSGDVADAAPGN N L G GAAPGN N L G SAAPG.....N.	368
Mhut 1271 AgSp1 cr	K DSE I IN K NGG F S N PFNFPG L NG F PGYICDYPDNG.. D ATTEGSQDLGGE I EGYGN G PFEDVADAAPEND L E.....	353

Varen 6869 AgSp1.1 cr		263
Varen 6869 AgSp1.2 cr	P	359
Mg TRINITY DN1546 c0 g1 Cr	GKEVKDNRNGSSGDVGNSFPDYS I VGVTPL T NQS....SGA	339
Mraby 7048 AgSp1 cr	 L GS S APG..N..... T LGS S AP	394
Ncruc 58532and13532 AgSp1 cr	LGGAAPD L GAPAPG..NDLGGAAPDLGAPAP G ND L GGAAPDLGAPAP..GNDLGGAADLG.....APAPAPGND L GATDPP S DQ..GSNTLVA	459
Msag 19329 AgSp1.1 cr		249
Msag 18528 AgSp1.2 cr	VGGAAPNEDGANDGGSQDLGGEV.KSNKVG P FRD V GGAAPYEDGANDGDSQDLGGEVKGSGNGPSGDVADAAPDDN M ADAPQ S NL..SSPTPVT	456
Lcorn 51676 AgSp1.1Cfrag cr	DLGSPA.....PENDLG G VAP.....NLGSPAPQ T NQG..SNTLVA	398
Lcorn 58650 AgSp1.2 cr	DLGSPA.....PENDLG G VAP.....NLGSPASE T NQG..SNTLVA	398
Amarm 50219 AgSp1.1 cr	DLGGA.....PENDLGGAAP.....ENDLGGAAPQ S EMG..SNTLVA	400
Aarg AgSp1 FC511 cr		368
Atri AgSp1 genomic cr	N L GSAA.....PGNN L GSAA.....ENDLGGAAPENNPAPQ S DT P SS	406
Mhut 1271 AgSp1 cr	GVASQ S NLG..PNTLVA	368

Varen 6869 AgSp1.1 cr	.NDPSCGDDVNSAFST A KQALQ	284
Varen 6869 AgSp1.2 cr	TSVTLCEDNVVSTF S V A KQALQ	381
Mg TRINITY DN1546 c0 g1 Cr	DID G NCDNDVATT F NT T K Q A I R	361
Mraby 7048 AgSp1 cr	PEDEGCQDNVPEAF S KARQALA	416
Ncruc 58532and13532 AgSp1 cr	PDDEDCQDNVPEAF H KARQALA	481
Msag 19329 AgSp1.1 cr		249
Msag 18528 AgSp1.2 cr	GT E GN C EDDVDRTLNT A KQALQ	478
Lcorn 51676 AgSp1.1Cfrag cr	PDDEDCQDNVPET F SKAR E ALA	420
Lcorn 58650 AgSp1.2 cr	PDNEDCQDNVPET F SKAR E ALA	420
Amarm 50219 AgSp1.1 cr	PEDEDCQDNVPEAF N KARQAAT	422
Aarg AgSp1 FC511 cr	PTDQ D CEED V HGA F SKARQALM	390
Atri AgSp1 genomic cr	PTDKGCEED V QGA F SKARQALM	428
Mhut 1271 AgSp1 cr	PEVK D CQDS V PEAF R KARHALT	390

☒ non-conserved
☒ similar
☒ ≥ 50% conserved